

What would prove it wrong

The literature review went to peer review and came back with one criticism that lands. This is what was said, what it changes, and the experiment register that replaces the ranking write-up.

Reviewed Session 29, week 15

Corpus 58 papers, 56 extracted

Deadline 30 September 2026

Fifth of five in this series

Five things were said

Ranked by how much each one changes. The first is the only criticism aimed at the method rather than the findings, and it is right.

“I feel that you are trying to justify your hypothesis with papers. New papers ideally should be giving birth to new hypotheses or experiments.”

CORRECT

The hypotheses were fixed before the corpus was gathered, and every judgement column measures fit to one of them. There was no cell in which a paper could propose something new. Fixed by a backward pass over data already collected.

“How can you say that H3 and H5 are unclaimed?”

FAIR AS STATED

The evidence is thin coverage — 6 and 7 papers — not absence. And the corpus is known to under-sample this benchmark.

The claim is downgraded to *not found in this corpus*.

“I frankly think that your hypothesis five and six are a single hypothesis.”

HALF RIGHT

Separable as mechanisms, one as an experiment. Merged into a single treatment with two ablation switches, which removes the H5-versus-H6 choice the session spent time on.

“That experiment should clearly state: this hypothesis is falsified if this happens.”

ADOPTED WHOLESALE

H1–H6 had mechanisms and metrics but no stop conditions and no baseline. The register below supplies both, and supersedes the planned hypothesis ranking.

“How do people start to understand the world? Psychology, cybernetics. I will not just look from the technical perspective.”

THE REAL GAP

The five clusters came from humans playing the game, then were answered entirely from machine-learning papers. The field that studies the humans was never opened.

The pipeline could not change its mind

Laid out as a sequence, the defect is visible in one line. Everything to the right of the marked node could only ever be sorted into what sat at it.

peer transcript

→

5 clusters

→

11 seeds

→

H1–H6 fixed

→

590 candidates

maps_to_H · verdict · cluster_final – three judgement columns, all asking "how does this relate to something I already decided". No column could carry a seventh hypothesis, so none

could ever appear.

What this does not invalidate

The corpus and the findings stand. The 58 were selected against the *clusters*, which came from humans playing the game rather than from H1–H6. The nine extraction columns are descriptive, written without reference to any hypothesis. And three of the review's main findings — the protocol confound, the open-weight count, and Rodionov's masking result — landed *against* prior expectation, which is not what a self-confirming review produces. The defect is narrower than bias: the review had no output channel for a hypothesis it did not start with.

The fix, in one pass

Nine columns in `shortlist_extracted.csv` are neutral — Mechanism, Agent state, Update trigger, Action efficiency, Stated limitations and four more. They describe what each paper does on its own terms. The pass re-reads only those, 58 times, asking the inverted question:

This paper's mechanism, described on its own terms — is it something H1–H6 names? If not, what is it?

One new column, `suggests_experiment`. Most rows will answer "covered by H4" or similar; a handful will not, and those are the point. `Stated limitations` is the richest source, because it is where authors record what they could not make work — a list of open problems written by people who tried.

Cost: one session. No new PDFs, no new extraction credits. The data was already paid for; it was simply never queried in this direction.

The frozen model is the method, not the constraint

The session settled on freezing an open-weight model and testing the harness on top. That was reached as engineering practice. The literature says it is also the only regime in

which the answer means anything.

Rodionov's ablation took the ARC-AGI-3 reference architecture and switched its parts off one at a time — world model, verification, simplification — to see which part carried the result. Scores ranged from 34.16 to 74.78 RHAЕ. The finding that matters is not the range but its cause: **the spread from changing the model and the reasoning effort exceeded the spread from changing the architecture.** A weaker variant on a stronger model beat a stronger variant on a weaker one.

REGIME	WHAT SITS ON TOP OF YOUR DELTA	WHAT YOU CAN CONCLUDE
Frontier model + harness	the model's own reasoning already performs much of what the harness performs, as a large uncontrolled term	a 2-point move is unattributable — harness, or noise
Frozen 7B + harness	the model demonstrably cannot hold the trajectory, notice a revisit, or remember across a reset	if the score moves, the harness moved it

So the Kaggle rule — no internet, local weights only — does not push the project into a worse experiment. It pushes it into the only experiment that can answer the question being asked. Two independent routes, the session's and the literature's, arrive at the same design.

Stated in advance: it is possible the harness helps only small models and a frontier model has no use for it. That is a real result, not a failure — but it must be predicted now rather than discovered in review.

The literature that was never opened

Every one of the 590 candidates came from citation hops out of machine-learning seeds. The cluster questions came from people playing the game. Human questions, answered exclusively from machine literature.

SOURCE FIELD	WHAT IT ESTABLISHED	CLUSTER IT ANSWERS
Developmental psychology	How infants approach an unfamiliar object: systematic single-variable variation, with play measurably closer to optimal	A — unknown world

SOURCE FIELD	WHAT IT ESTABLISHED	CLUSTER IT ANSWERS
Gopnik, Spelke	information-gathering than to random poking	B — probably true
Affordances J. J. Gibson	Perception delivers actions, not object labels. You do not see a handle and infer pull-ability; you perceive pull-ability directly	D — knowledge into action
Cybernetics W. R. Ashby	Requisite variety: a controller needs at least as many distinct responses as the environment has distinct states, or control is provably impossible	C — discovering information
Perceptual control W. T. Powers	Organisms control their perceptions, not their actions, varying action to hold a perception steady — the standard loop inverted	E — improving over experience

These are the fields where the cluster questions were first posed, decades before anyone built an agent for them. Each is likely to return either a mechanism the machine-learning corpus has not tried, or a clean argument for why a planned one will not work. Gibson is already the second kind: **affordance theory is a live threat to H3**, which assumes controllability is a separate property to be identified in a dedicated phase. Worth knowing before that experiment is built.

Bounded deliberately: **15–20 sources, review articles and textbook chapters, one sitting**. One cluster question at a time. It does not enter the 58, gets no extraction, and does not reopen the funnel — it is filed as a separate appendix.

The experiment register

Every experiment is written here before it is run, with its falsifier stated in advance. An experiment with no falsifier is not an experiment; it is a hope. A block may be edited freely until its first run — after that, changes are appended as dated amendments, never made in place.

Fixed across all experiments

Model	One frozen open-weight SLM — Qwen 2.5 7B or Gemma 3, decided empirically at E0. No we in this register.
Independent variable	The harness, and only the harness. Identical weights, decoding parameters and prompt scaling treatment.

Split	25 public games / 183 levels, split once at E0 into 8 tune and 17 held-out. Every number be from held-out.
Seeds	N = 5 minimum. A delta smaller than the baseline's seed-to-seed spre
Reporting	A vector over clusters A–E, never a single score.

Why a vector, not a score

A harness that promotes failures into rules may suppress exploration: a gain in E paid for with a loss in C, invisible under one RHAE number. So metrics.py emits a proxy per cluster on every run — levels reached on first contact (A), rate of acting on a later-contradicted belief (B), distinct states per 100 actions (C), actions between an effect being learned and used (D), repeat-hazard rate after first failure (E). A regression outside the target cluster does not sink an experiment. It is recorded, and it is the thing the write-up has to explain.

E0

Baseline characterisation

BLOCKED — GPU ACCESS

QUESTION	Does a frozen open-weight SLM with a minimal scaffold score anything measurable on ARC-AGI-3?
SETUP	Observe → choose action → observe. No memory beyond the raw context window. No harness.
MEASURED	RHAE, levels reached, actions-to-solve, the five cluster proxies, and seed-to-seed variance on all of them.
FALSIFIED IF	RHAE is 0.0000 on every held-out game across all five seeds — the baseline indistinguishable from the published random-agent floor.

IF FALSIFIED

The register does not stop. RHAЕ only registers a level actually solved. Switch the primary metric to dense alternatives that move before any level completes — levels reached, distinct states visited, repeat-hazard rate — and report RHAЕ secondary. A harness can be demonstrably better while both conditions still score zero officially, and that must stay visible rather than be fatal.

ALSO DECIDED

Which of the two candidate models, on floor and variance. The tune / held-out split, fixed here and never touched again.

E1 Trajectory memory — H5 and H6 as one treatment

NEXT AFTER E0

HYPOTHESIS

An agent that recognises a return to a visited state and prunes that branch, and converts a failure episode into a rule surviving the reset, solves levels in fewer actions and dies the same death less often than the same model without those stores.

SWITCHES

`--prune-visited` (H5) and `--failure-rules` (H6). Four cells: off/off is E0, then on/off, off/on, on/on.

PRIMARY

Actions-to-solve on held-out games.

FALSIFIED IF

on/on does not beat off/off by more than the baseline seed spread, on held-out, at N = 5.

SPLIT VERDICT

If exactly one switch carries the effect, the two mechanisms are not one experiment after all — amend the register to say which, and why the merge was wrong.

ABANDONED IF

Beyond 40 GPU-hours, or the delta is still inside noise at N = 10.

PRIOR ART TO BEAT

PRO-LONG argues the opposite of H6 — retain the complete log and search it with code rather than consolidating it into rules. A null on `--failure-rules` is *evidence for PRO-LONG*, and gets written up that way rather than as a dead end.

E2 Controllability probe — H3

RUNS ONLY IF E0 YIELDS A FLOOR

HYPOTHESIS	Spending the first k actions identifying which on-screen object responds to input lowers total actions to solve, relative to spending those same k actions on the task.
CONTROLS	E0, <i>and</i> a matched-budget control — E0 given the same k extra actions with no probe instruction. This separates "the probe helped" from "more actions helped".
SWEEP	$k \in \{5, 10, 20\}$, on tune games only, fixed before held-out is touched.
FALSIFIED IF	The probe does not beat the matched-budget control outside seed spread. Beating plain E0 while losing to the matched control means the probe bought nothing — the extra actions did.
LIVE THREAT	Gibson's affordance account holds controllability is perceived in the act of looking, not established in a separate phase. If E2 nulls, that is the likelier explanation than a bad implementation — and Tycho's finding that high transition-model accuracy did not translate into action-selection gains points the same way.
COVERAGE CAVEAT	H3 rests on the thinnest support in the shortlist — 6 papers, in a corpus known to under-sample this benchmark. <i>Not found here, never unclaimed in the field.</i>

E3 Epistemic versus instrumental labelling — H2

PARKED

WHY PARKED	H2 is claimed and named on this benchmark. Tycho calls it "active abstraction"; OPINE-World supplies a quantified signal in Bayesian ontology error steering exploration. Reproducing a claimed mechanism costs the same compute as testing an open one and returns less.
UNPARKED IF	E1 and E2 both null, and a mechanism with published traction is needed to establish the harness works at all.

Register-level stop conditions

CONDITION	ACTION
E0 falsified and no dense metric separates conditions either	Stop. Report the floor problem as the finding — a negative result with a clean baseline is a milestone deliverable, not a failed milestone.
Deadline within seven days	Freeze the register, write up everything holding a verdict, mark the rest "not run" with the reason.
An experiment beats its falsifier	Do not build on it yet. Re-run at N = 10 on held-out first. One surviving delta at N = 5 is a candidate, not a result.

Where the work stands

The literature review is closed. The session's own ceiling — "maybe one hundred papers, then come up with a set of experiments" — puts 58 papers and 56 extractions comfortably inside. Everything that adds reading from here is time-boxed or does not happen.

01	Experiment register, live Supersedes the planned hypothesis ranking	NEW · DRAFTED
02	Kaggle account and phone verification Gates GPU access, and therefore E0 and everything downstream	BLOCKING
03	Model decision Frozen open-weight SLM, hosted inference during development, weights uploaded for submission	CLOSED
04	Cluster-vector metrics in <code>metrics.py</code> Reframed from the scalar efficiency ratio carried since 2 September	REFRAMED
05	Backward pass over the 58 One new column, one session, no new reading	NEW

- 06

Cross-domain pass

15–20 sources, one per cluster question, filed separately

NEW · BOUNDED
-
- 07

Downgrade the H3 / H5 "unclaimed" claim

One edit to the milestone brief, one to the findings note

NEW
-
- 08

Research-harness skill file

The pipeline is already scripted; what is missing is the wrapper that runs it on someone else's question

NEW · FOR PEERS
-
- 09

Worked ARC examples in the findings note

The findings read as claims about literature; they should read as claims about the game

NEW
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On what gets kept. "I do not care whether you are top of the leaderboard. I want to see what you have done behind." The failed co-citation rule, the citation-graph blind spot, the palette that failed its own accessibility check, and now a review that could not change its mind — these stay in the record where they happened. The register exists so the next set of mistakes is as legible as the last.

Fifth note in the ARC Prize 2026 Milestone #2 series, after the milestone brief, the hypothesis plan, *From 590 to 58* and *What the 58 Say*. Sources: Session 29 transcript, week 15; the 58-paper shortlist with full-text extraction across 56; Rodionov 2607.15439; PRO-LONG 2607.20064; OPINE-World 2607.01531. Compiled 10 September 2026.